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COMMENT

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Introduction

The Swedish Society of Infectious Diseases expert group for pneumonia has provided evidence-based guidelines for the management of adult immunocompetent patients with community-acquired pneumonia (CAP) since 2005. The current guidelines are the third update and are based on a systematic literature search in of MEDLINE for publications in between September 2015 and June 2024 and limited to articles published in English with available abstracts. The strength of the recommendations is based on the Infectious Diseases Society of America's (IDSA) evidence grading system [1].

The guidelines apply to the management of non-immunocompromised patients with CAP that seek hospital care. Pneumonia is defined as an infection of the lung parenchyma, characterised by symptoms and clinical findings indicative of an acute lower respiratory tract infection, accompanied by radiological changes. Common symptoms include fever, cough, dyspnoea, fatigue and chest pain. Pneumonia is classified as community-acquired when it is contracted outside of healthcare settings. However, the definition lacks clarity, and the time frame varies across studies. It is crucial to distinguish between bacterial CAP and hospital-acquired/healthcare-associated pneumonia (HAP) due to their differing in aetiological spectra. Nonetheless, many aspects of these recommendations can also be applied to HAP. The guidelines do not encompass detailed management of COVID-19, pneumonia with sepsis/septic shock or details regarding vaccinations as these topics are covered in other Swedish national guidelines. All guidelines can be accessed from the Swedish Society of Infectious Disesases website [2].

Incidence and mortality

The annual incidence of CAP in adults requiring hospitalisation in high-income countries has been estimated to approximately 300/100,000 [3–5]. Several studies indicate a gradual increase, especially among the elderly [4–6]. The incidence is slightly higher in men compared to women [3,7]. The true incidence is, however, uncertain due to potential misclassification. An 2024 American study involving 17,000 hospitalised patients with CAP found that the diagnosis was incorrect in 12% of cases, particularly among elderly individuals with cognitive impairments [8]. Mortality from pneumonia varies depending on age and underlying health conditions [3]. A Swedish registry study of all hospitalised patients with pneumonia as the primary diagnosis between 2005 and 2015 reported a 30-day mortality rate of approximately 9% [4]. For patients registered in the Swedish Pneumonia Registry from 2008 to 2023, the in-hospital mortality was 4.4% based on approximately 57,000 hospitalisations, with a higher risk in men (5,2 compared to 3,6%) in women ($p < 0.01$). Previous studies have shown varying mortality risks, some reporting higher mortality in men [7,9], while others have shown equivalent or higher mortality in women [3,10]. Patients hospitalised for pneumonia also exhibit higher long-term mortality risk compared to age-matched controls [11–15]. However, it remains unclear whether the increased long-term mortality is attributable to the infection itself or if the diagnosis serves as a marker of underlying morbidity [15]. Pneumonia is also associated with increased morbidity from other diseases, both in the short and long term, such as increased risk of cardiovascular disease [16].

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Aetiology

CAP can be caused by a variety of microorganisms. Determining the aetiology can be challenging, especially in patients who have received antibiotic treatment prior to microbiological sampling [17] or have difficulty providing sputum samples [18]. In the Swedish Pneumonia Registry (data from 2008 to 2023), a causing pathogen was identified in 49% of cases. The most frequent aetiology was *Streptococcus pneumoniae* (14% in total of all episodes) followed by *Hemophilus influenzae* (9%), *Mycoplasma pneumoniae* (5%), influenza virus (3%), *Staphylococcus aureus* (1%), *Legionella pneumophila* (1%) and *Chlamydia pneumoniae* (0.2%). At least two pathogens were detected in 1.5% of patients, of which *S. pneumoniae* was identified in almost half of these cases. For patients treated in the ICU for pneumonia, the proportion of *S. pneumoniae* was 24%. Even in patients treated for CAP in European ICU's, bacterial findings are dominated by *S. pneumoniae*. Less common agents reported include *S. aureus* and *Legionella spp* [19]. Aetiological studies are difficult to compare due to variations in study populations and diagnostic methods. A meta-analysis by Said et al. 2013 detected *S. pneumoniae* in 5–51% by either blood culture, sputum culture and/or urine antigen test [20]. In more recent publications, the microbiological aetiology varies [18,21,22]. In two Swedish studies from 2010 where extensive diagnostics were performed (culture, PCR and urine antigen test), *S. pneumoniae* was detected in 48–63% [17,23], while another Swedish study from 2023 found that *S. pneumoniae* and viral aetiologies each accounted for 28% and *H. influenzae* for 16% of cases. In the same study, the respective agents were detected in 6%, 7% and 7% in respiratory healthy controls [24].

In contrast, a 2021 Danish study reported a dominance of *H. influenzae* [25] and a 2024 Norwegian study found *H. influenzae* to be predominant in hospitalised pneumonia patients [26]. Differences compared to the Swedish studies may be due to a higher proportion of samples collected from the lower respiratory tract and improved diagnostic methods. Among patients with chronic obstructive pulmonary disease (COPD), *H. influenzae is* the most common bacterial finding in exacerbations [27], but *S. pneumoniae* remains the predominant pathogen causing pneumonia [28,29]. Approximately half of COPD exacerbations are caused by respiratory viruses [30].

The frequency of respiratory tract infections caused by *M. pneumonia, C. pneumoniae* and *Chlamydia psittaci* varies over time. Epidemics with *M. pneumoniae* often occurs at 4–7 year intervals [31]. The seroprevalence of *C. pneumoniae* is high in the population, and high antibody titres may be detected in healthy individuals [32–34]. Despite this, *C. pneumoniae* is uncommon among hospitalised patients in Sweden [17,23] and its pathogenicity has been questioned [35] but outbreaks have been reported [36–38]. Infections caused by *C. psittaci* are likely underdiagnosed [39]. A retrospective study from the Netherlands found a frequency of 4.7% in patients with CAP by combining PCR with antibody testing [40]. Seasonal variations and local clustering of cases have been described and are often related to bird handling [41,42]. In Sweden, the majority of cases are diagnosed during the winter season [43].

Viral aetiology is common among both children and adults with pneumonia [44,45] and has gained increased attention since the expanded use of diagnostic PCR was introduced [46,47]. Multiple microorganisms may be detected [17,47,48] and mixed infections with viruses and bacteria can be associated with severe pneumonia [44,49,50]. The clinical significance of PCR-detected respiratory virus in a patient with pneumonia can be difficult to assess. A single finding of influenza virus, respiratory syncytial virus (RSV), human metapneumovirus, rhinovirus or SARS CoV-2 in an adult patient with CAP is however likely to be clinically relevant [22,45,51,52].

The causative pathogen in CAP cannot reliably be predicted based solely on clinical and/or radiological findings [53–56]. Elderly patients (>75 years), often present with unspecific symptoms and may lack fever [57]. Acute onset of symptoms, pleuritic chest pain and a WBC $>15 \times 10$/L may suggest pneumococcal aetiology [58–61]. The classic clinical picture of infection with atypical pathogens (*M. pneumoniae, Legionella spp* and *Chlamydia spp*) is characterised by younger age, gradual onset, dry cough and white blood cell (WBC) count $< 10 \times 10^9$/L [53,62–64], while the combination of sudden onset of confusion, liver dysfunction, hyponatremia and relative bradycardia increases the likelihood of *Legionella spp* [63,65,66]. *Bordetella pertussis* generally causes a pro-longed whooping cough in adults which may pave the way for secondary infections. In recent years, a marked increase of reported cases of *B. pertussis* has been noted in several European countries, including Sweden [43]. The incubation

period can provide guidance on the causative agent and is typically 2–3 weeks for *M. pneumoniae* [67], 2–10 days for *Legionella spp* [68] and 1–3 days for influenza [69]. Lack of clinical improvement for patients treated with oral penicillin may increase the suspicion of atypical pathogens but does not exclude pneumococcal aetiology [59,65]. Geographical differences in prevalence and epidemiological data should be considered, especially when the patient has been infected during travel.

Transmission

The scientific evidence regarding the transmission dynamics of respiratory infections is generally of low quality [70]. The minimum infectious dose is often unknown, and the risk of infection may vary due to several factors [71]. For bacterial pneumonia caused by *S. pneumoniae*, *H. influenzae*, *M. catarrhalis* or *S. aureus,* infection is typically preceded by colonisation of the upper respiratory tract followed by micro aspiration [72], which differs from CAP caused by atypical bacteria or respiratory viruses. Respiratory infections are primarily considered to spread *via* droplets of various sizes, though indirect and direct contact transmission also occurs. Traditional infection prevention and control (IPC) measures for respiratory infections, which a strict separation between droplet and airborne transmission and accordingly between adjacent recommendations for specific personal protective equipment have recently been questioned [73–75]. A 2024 report by the World Health organisation (WHO) proposes a new terminology, simplifying the classification into airborne/inhalation transmission and direct/indirect contact transmission [76]. It is crucial to note that the risk of transmission from an individual with pneumonia cannot be assessed solely based on microbiological findings; it also depends on other factors such as the duration of illness, severity of symptoms, presence of immunosuppression and environmental conditions [73,77]. Access to rapid and accurate diagnostics in emergency departments can mitigate the risk of in-hospital transmission and facilitate the prioritisation of single-room care [78–80]. Patients admitted for other reasons may have an unknown previous exposure and might develop symptoms during hospital stay. Additionally, staff and visitors can serve as a potential source of infection [81,82]. Therefore, microbiological testing should be performed liberally on admission as well as during hospital stay for patients with respiratory symptoms in order to prevent hospital outbreaks.

Bacteria that cause pneumonia are primarily spread through close contact (direct or indirect) with respiratory secretions. Carriage of *S. pneumoniae, H. influenzae* but also *Streptococcus pyogenes* and *M. catarrhalis* are common among children and do not necessarily lead to symptomatic infection in adults. For atypical pathogens, carriage is uncommon, but PCR positivity may persist for several weeks after the actual infection. Adequate antibiotic treatment quickly reduces bacterial amount and thus the risk of spreading infection. In clinical practice, the risk of transmission is often considered negligible 48 h after clinical improvement/resolution of fever in bacterial pneumonia. The infectivity of *M. pneumoniae* is high and outbreaks have been described in close contact settings such as schools and military facilities but also in healthcare facilities [83–85]. Assessing the duration of infectiousness is challenging due to the PCR-based diagnostics, the high proportion of mild infections and the long incubation period. For *C psittaci*, person-to-person transmission is considered rare but has been reported [86,87]. For *L. pneumophila*, however, only a single case of person-to-person transmission has been described [88].

Shedding of respiratory viruses may be high early in the disease course [89] even in patients with mild symptoms. Virus can typically be detected by PCR for a few weeks in immunocompetent individuals, where rhinoviruses persists for the longest period [90]. A 2019 Swedish study of 444 adults without reported respiratory symptoms detected respiratory viruses in 4.3%, predominantly rhinoviruses [91]. In-hospital transmission of influenza, RSV and SARS-CoV-2 may have serious consequences [81,92,93]. For patients with suspected infectious respiratory viruses, IDSA and European Centre for Disease Control (ECDC) therefore recommend single room care or co-horting with patients with the same verified infection [94,95]. Individuals with other types of respiratory viruses can also pose a risk to fellow patients [96,97]. If a hospitalised patient has been exposed to influenza virus, early initiation of antiviral prophylaxis is recommended as the majority of hospitalised patient can be considered to belong to a risk group for severe influenza [98]. Typically, a patient is deemed non-infectious no earlier than 5 days after symptom-onset and $>$24 h of fever resolution, or after three days of antiviral therapy. In

immunocompromised patients, high viral loads can persist for significantly longer time periods [99], which is why an individual assessment of infectiousness should be made in such cases.

Recommendation

- Patients with respiratory infections requiring hospitalisation should undergo diagnostic testing and if suspected or verified infectious be cared for in a private room (or in a shared room if fellow patients have the same infection) *(C III)*.

Antibiotic resistance

In 2024, the European Committee on Antimicrobials Susceptibility Testing (EUCAST) established new breakpoints for benzylpenicillin in *S. pneumoniae* with MIC $S \leq 0.06$ mg/L, $I > 0.06 - 1$ mg/L and $R > 1$ mg/L [100]. In Sweden, the proportion of blood isolates with *S. pneumoniae* with PcG I/R has ranged between 5 and 8% and the incidence of notifiable strains (MIC >1 mg/L) between 0.6–1.4 cases per 100,000 inhabitants since 2015 [101], which is low from a European perspective [102]. The proportion of invasive pneumococcal isolates resistant to other relevant antibiotics has also been stable: for clindamycin, 4–7%; erythromycin, 4–7%; trimethoprim -sulfamethoxazole, 2–5%; tetracycline, 4–9%; fluoroquinolone, 0–1%. The proportion of *H. influenzae* with β-lactam resistance has remained stable at around 35–45% over the past 10 years [101], amoxicillin-resistant isolates ranged between 25% and 35%, while resistance to trimethoprim-sulfamethoxazole has ranged between 25% and 30%. It is, however, unclear whether these proportions are representative for patients with CAP [103].

Macrolide resistance in *M. pneumoniae* has been observed since 2000 but remain uncommon in Sweden. In a Swedish study reporting data from 1996 to 2013, no resistant strains could be identified [104] in contrast to other parts of Europe and the USA where the proportion of macrolide-resistant strains is around 10% [105–107]. In Japan, it has been found that over 80% of *M. pneumoniae* isolates have a point mutation leading to macrolide resistance, and in China the prevalence is 69–100% [108]. In rare cases, *M. pneumoniae* may develop resistance during ongoing macrolide treatment [109,110].

Severity assessment

Various assessment tools have been utilised to evaluate the severity of infections. The National Early Warning Score 2 (NEWS2) effectively predicts the need for intensive care, whereas pneumonia-specific assessment tools such as the Pneumonia Severity Index (PSI) and CRB65 predict mortality more accurately [111]. Two meta-analyses comparing mortality associated with pneumonia found CRB65 to be equivalent to the more complex PSI and CURB-65 [112,113]. A large prospective study conducted 2014–2017 demonstrated lower mortality in patients assessed with CRB65 or CURB65 compared to those where no scoring was performed [114]. However, a limitation of CRB65 is its precision in identifying patients with severe pneumonia. A German study involving over 300,000 hospitalised pneumonia patients reported an overall mortality rate of 14% and 2.4% for patients who scored 0 points by CRB65 [3].

Combining CRB65 with D (denoting other serious diagnoses) and S (denoting saturation $<$90%), expressed as DS-CRB65, has been evaluated in several studies, including two from Sweden. DS-CRB65 demonstrates better sensitivity in predicting severe outcomes, such as the need for intensive care or mortality, compared to CRB65 [115–117]. However, an American retrospective study found no difference in the ability to predict mortality in pneumonia when DS-CRB65 was compared with qSOFA, SIRS, CURB and CRB65 [118]. Apart from statistical performance, it is crucial that an assessment tool is educational and user-friendly, why DS-CRB65 is recommended in these updated guidelines. It is important to emphasise that all scoring systems should be used as a support for initial assessment and management and need to be supplemented with an evaluation of individual patient factors and laboratory findings, see Figure 1.

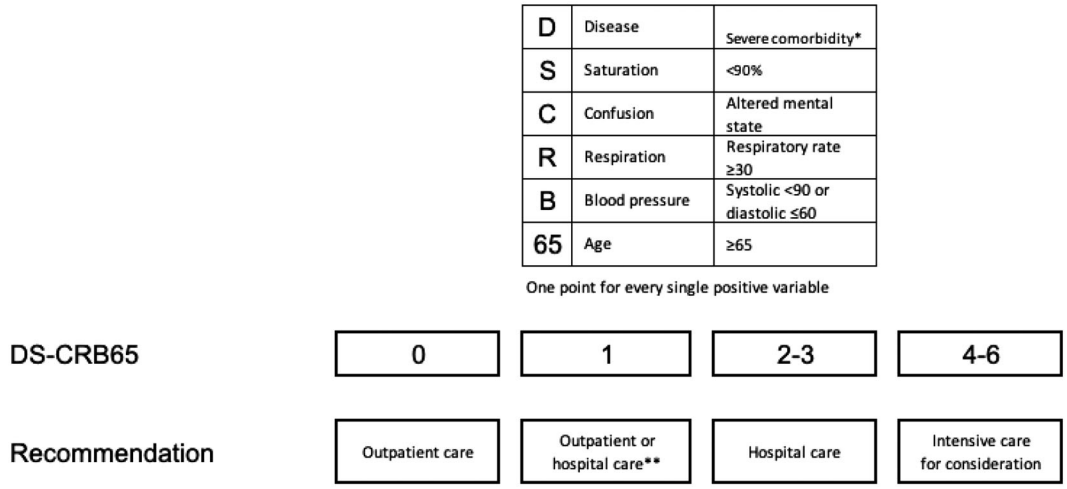


Figure 1. DS-CRB65 To support decisions about level of care.
* Severe comorbidities refer to presence of at least one of following conditions: congestive heart failure, chronic kidney/liver/neurological/cerebrovascular disease, COPD or active malignancy.
** Hypoxia, high respiratory rate, low blood pressure or confusion indicate need for hospitalisation.

Recommendation

- Assess disease severity with the support of DS-CRB65 *(A II)*.
- Assess the appropriate level of care, microbiological diagnostics and empirical antibiotic treatment on an individual basis but with support from DS-CRB65 *(B II)*.
- Monitor severity during in-hospital care with NEWS2 *(A II)*.

Management of circulatory and respiratory failure

Circulatory failure associated with pneumonia does not significantly differ from other severe infections. In cases of severe pneumonia, close monitoring is warranted due to the frequent combination of respiratory and circulatory failure. A more restrictive fluid therapy strategy compared to previous practices has been suggested in recent years, although strong evidence is lacking regarding the most beneficial regimen for patients with severe CAP. Guidelines from the Surviving Sepsis Campaign 2021 indicate a lack of scientific support for a definitive recommendation on whether restrictive or liberal fluid therapy is preferable in initial care [119]. In septic hypotension, the administration of crystalloid fluid (at least 30 mL/kg) is recommended during the first 3 h followed by maintenance therapy when circulatory stability has been achieved. Management regarding vasopressor therapy does not differ for patients with CAP from other conditions with sepsis or septic shock.

Respiratory failure is common in severe pneumonia and is associated with increased mortality. Hypoxia can also lead to worsen outcomes in less severe pneumonia [120]. A target saturation of $\geq$93% should be maintained for most patients. Oxygen can be administered *via* nasal cannula up to 5 L/min or by mask up to 15 L/min, depending on the type of mask. For patients with a saturation $>$96%, oxygen should not be administered as studies have shown adverse effects of hyperoxia in acute illness [121]. High-flow nasal cannula oxygen (HFNC/O) delivery, a well-established method in neonatology, has recently been widely adopted for adults. High-flow systems offer several advantages over traditional methods, including the ability to deliver heated and humidified oxygen at flows up to 70 L/min with an oxygen fraction up to 100%. A randomised study of pneumonia patients found a 90-day mortality twice as high with conventional oxygen therapy compared to high-flow systems [122]. HFNC/O is preferred over conventional oxygen therapy in acute hypoxic respiratory failure, defined as PaO2/FiO2 $\leq$ 200 mmHg, according to recommendations from the European Respiratory Society (ERS) 2021 and reduces the risk of intubation [123].

Continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP) can be used for patients with moderate to severe pneumonia. Indications typically include hypoxia, secretion mobilisation and prophylaxis of atelectasis. BiPAP offers the advantage of providing respiratory support and carbon dioxide elimination in addition to hypoxia treatment. The availability outside the intensive care setting is increasing, which may benefit patients treated for CAP [124]. Non-invasive ventilation (NIV) is an option for selected groups. A 2020 meta-analysis noted a significant reduction in the risk of intubation in patients with pneumonia treated with NIV [125]. Two studies demonstrated faster reversal of hypoxia in those treated with CPAP compared to conventional oxygen therapy [124,126]. The ERS recommends that NIV can be considered for pneumonia patients without contraindications, provided that the treatment is performed by experienced personnel and with intubation readily available in case of deterioration [123]. In summary, there is clinical experience but insufficient scientific evidence to provide recommendations regarding the use of NIV in patients with CAP.

Recommendation

- Administer at least 30 mL/kg of isotonic crystalloid fluid within the first three hours in cases of hypotension or hypoperfusion according to sepsis criteria *(B II)*.
- Individualise fluid therapy once circulatory stability has been achieved *(B II)*.
- Aim for a saturation of $\geq$93% for most patients *(A II)*.
- Refrain from oxygen therapy if the saturation is $>$96% *(B II)*.
- Check arterial blood gases and consult with an intensive care specialist if difficulties in achieving adequate saturation *(C III)*.
- In case of acute hypoxic respiratory failure, administer oxygen via high-flow systems to reduce the risk of intubation *(A II)*.

Biomarkers

Inflammatory biomarkers can support decisions about antibiotic initiation/discontinuation, prognosis, level of care and hospital discharge [127,128]. Furthermore, the potential of inflammatory biomarkers to distinguish between bacterial and viral aetiology has been investigated. The dynamic nature of these biomarkers however presents methodological challenges in study design and interpretation as patients are at different stages of infection upon hospital arrival [129]. Elevated levels of cardiac biomarkers can be associated with both the infection itself and an increased risk of cardiovascular complications within 30 days [130].

C-reactive protein (CRP)

C-reactive protein (CRP) is synthesised in the liver and is present at low levels ($<$3 mg/L) in healthy individuals but the concentrations rise due to cytokine stimulation associated with infection or inflammation. Studies have shown that CRP is an independent marker of severe disease both initially and later in the course [131,132]. Pneumonia caused by *S. pneumoniae* and *Legionella spp* often induce a sharp increase in CRP levels [59,133,134], while atypical agents and viral infections generally induce lower CRP levels. However, CRP cannot be used to discriminate between bacterial and viral aetiology in general [135] and is often lower in the elderly [135–138]. The kinetics of CRP can indicate the prognosis of pneumonia. Hospitalised patients whose CRP do not decrease by at least 50% after four days of treatment are more likely to require mechanical ventilation, vasopressor therapy and had an increased 30-day mortality [112]. Other studies have reported an association between a slowly decreasing CRP and 30-day mortality [139,140]. CRP levels may also correlate with the extent of radiological findings [141].

Procalcitonin (PCT)

Procalcitonin (PCT) is a precursor to the thyroid hormone calcitonin and is found at very low levels in serum of healthy individuals [142]. Secretion is stimulated by various inflammatory and infectious

conditions. In recent years, PCT has been increasingly used for diagnostic, prognostic and even therapeutic considerations in pneumonia [143]. In a Swedish study of hospitalised patients with CAP, high levels of PCT on hospital arrival were associated with positive blood culture findings of *S. pneumoniae* as well as pneumococcal infection without concomitant bacteraemia [133]. Algorithms for antibiotic treatment based on serum PCT levels have been suggested in several studies [143–145]. A systematic review included 26 studies and 6,708 patients with acute respiratory tract infections treated with antibiotics either based on a PCT algorithm or according to standard care [144]. In the overall population, mortality was significantly lower when the PCT algorithm was applied but not among the subgroup with CAP. No difference in length of hospital stay could be demonstrated. A shortened duration of treatment was noted with the PCT algorithm (7,5 days vs. 10,4 in the control group) but other studies have not been demonstrated any reduction in treatment duration with similar PCT algorithms [146,147].

CRP and PCT have shown comparable ability to predict mortality within 28 days, if analysed on arrival and combined with a severity assessment [129]. A retrospective study of hospitalised elderly patients showed higher accuracy for CRP than PCT in identifying pneumonia [148], it is not certain which marker is of greatest value [149]. In the 2023 guidelines from the ERS, European Society of Intensive Care Medicine (ESICM), European Society of Clinical Microbiology and Infectious Diseases (ESCMID) and Latin American Thoracic Association (ALAT), PCT is recommended to monitor the course of patients with severe pneumonia in the ICU with the aim of reducing the number of days of antibiotic treatment [150]. However, we recommend CRP above PCT due to its lower cost and more established use in Sweden.

Neutrophil-lymphocyte ratio (NL-ratio)

The neutrophil/lymphocyte ratio (NL-ratio) can be used to assess the likelihood of severe bacterial infection in the acute phase [151]. A prospective study of 395 pneumonia patients showed that the NL ratio predicted mortality better than CRP [152] and in addition, a systematic review showed that an NL ratio $\geq$10 predicts mortality better than CRP, WBC, neutrophil count, PCT, PSI and CURB-65 [153]. Combining the NL-ratio with CURB65 further improves accuracy [154,155]. Another advantage is the ability to identify patients early in the disease course where CRP may be low despite severe disease.

Lactate

Elevated lactate is an established marker of hypoperfusion and poor prognosis in bacterial infections. Monitoring lactate levels is important in evaluating initial treatment. Combining lactate with severity assessment can improve the reliability of a prognostic assessment [156]. Arterial blood gas analysis is recommended in cases of suspected sepsis/septic shock and/or respiratory failure. An elevated lactate level is a stronger negative prognostic factor than hypotension in sepsis [157–159]. Therefore, lactate is a valuable addition to assess the severity of the infection.

Recommended sampling (C III)

- CRP
- Blood status (HGB, WBC, thrombocytes and differential count or N/L ratio)
- Electrolyte status (Na, K and creatinine)
- Liver status (AST, ALT)
- Lactate (venous or arterial)
- Arterial blood gas (if respiratory failure)

Microbiological diagnostics

Bacterial cultures

In addition to blood cultures, sputum samples for culture and susceptibility testing are recommended for all patients capable of expectoration to enhance the likelihood of identifying the aetiological agent

[160–163]. Induced sputum samples can be obtained by inhalation of NaCl [163–165]. While some laboratories assess the representativeness of sputum samples, the value of this practice has been questioned [166]. Findings of bacteria from non-representative samples may also be clinically relevant [167]. Nasopharyngeal sampling should be performed in all patients who are unable to expectorate. Findings of *S. pneumoniae* and *H. influenzae* in nasopharyngeal sample may be significant [168–170]. In cases of suspected *Legionella spp* lower respiratory tract specimens should always be obtained for both PCR and culture. Bronchoscopy should be considered for patients with severe pneumonia and those unresponsive to antibiotic therapy [171,172].

Pneumococcal antigen in urine

Rapid tests for the detecting of pneumococcal antigen in urine have a moderate sensitivity (70–75%) but high specificity (95–97%) for pneumococcal aetiology [173,174]. Sensitivity is higher in cases of bacteraemia. A negative test result does not exclude pneumococcal aetiology and should not prompt broader antibiotic therapy. The test can support pneumococcal aetiology when other microbiological examinations are negative [175–177]. Swedish studies report similar results [17,178–180]. Positivity can persist for several weeks after a past infection (including otitis) complicating interpretation [181–183]. Swedish data suggest that a broad recommendation of testing does not reduce the use of broad-spectrum antibiotics [184]. The test should therefore primarily be considered in selected situations, such as for severely ill patients [173] or to enhance diagnostic accuracy [185,186].

Legionella antigen in urine

The rapid test for legionella antigen in urine has high specificity for infection caused by *L. pneumophila* serogroup 1, although false positives occur [187]. Sensitivity is moderate and varies with disease severity [188]. A Spanish study of patients with pneumonia caused by *Legionella* reported a sensitivity of 38% for mildly ill and 86% for severely ill patients [189]. Several other serogroups and species in the *Legionellae* family may cause pneumonia [190] why a negative test result should not be used for de-escalating antibiotic therapy. The test is useful in selected situations with clinical/epidemiological suspicion of *L. pneumophila* infection, such as recent travel abroad or severe pneumonia of unknown aetiology [138].

PCR for bacteria

PCR for atypical bacteria can provide support for a change in antibiotic regimen [191,192] and has replaced serological diagnostics [193–195]. PCR for *C. pneumoniae, C. psittaci* and *B. pertussis* is often available in Swedish laboratories. PCR for *Legionella spp* on lower respiratory tract secretions has been shown to improve diagnostics compared with culture and urine antigen testing [196,197]. Multiplex PCR which includes bacteria and viruses is now available in most laboratories. PCR for *S. pneumoniae* and *H. influenzae* is less clinical significant compared to conventional culture but may increase the diagnostic yield [198]. PCR on respiratory samples offers higher sensitivity compared to conventional culture and is less affected by prior antibiotic treatment [199–201]. An important consideration is that PCR on respiratory secretions can be positive even in the presence of non-viable bacteria and in asymptomatic carriers without clinical relevance. PCR on lower respiratory secretions for *Legionella spp* should be supplemented with culture to enable epidemiological investigations [197]. Some species, such as *L. bozemanae* and *L. longbeachae,* are rarely included in respiratory panels for PCR diagnostics, with the latter species causing several cases in Sweden in 2018 [202].

PCR for viruses

Diagnostic PCR should be used during the season for respiratory viruses and if the patient requires admission to support decisions of initiation of antiviral treatment/withdraw of antibiotics and to reduce the risk of healthcare-associated transmission. Some respiratory viruses exhibit a typical seasonality, with influenza and RSV causing epidemics in the winter, while others (e.g. rhinovirus and adenovirus) may

occur year-round [77]. It is also important to stay updated on the global occurrence of high-risk pathogens such as avian influenza H5N1 and Middle East Respiratory Syndrome (MERS) where sampling may be justified for surveillance. Samples from severely ill influenza patients should be sent to the Public Health Agency of Sweden for genetic characterisation. Most Swedish laboratories offer both specific and combined PCR analyses for several respiratory viruses, but the benefit is primarily seen in severely ill and immunosuppressed patients [203]. A disadvantage of PCR is the difficulty to separate non-viable virus from clinical infection. In the case of a negative test and a strong clinical suspicion, additional testing should be considered; deficiencies in sampling technique may occur and virus levels may be low early in the course of the disease [204]. The 'threshold cycle value' (Ct value) can be considered as a semi-quantitative measure of viral load. Along with the assessment of symptoms, Ct values can provide guidance in assessing the individual's infectiousness and estimate duration of the disease [204,205]. However, the significance of the Ct value varies between different agents and is not yet standardised. In some cases, virus can be detected in samples taken from the lower respiratory tract even if samples from the upper respiratory tract are negative [206,207]. Table 1 outlines the recommended microbiological sampling based on severity assessment.

Radiological examinations

Traditionally, radiological diagnostics have referred to pneumonic *infiltrates*. Given that infiltrate is difficult to define [94,208], the preferred nomenclature for conventional chest X-ray and computed tomography (CT) is now instead *opacity* [209]. The most common radiographic findings in pneumonia are consolidations and ground-glass changes. The latter involve densifications of the parenchyma which, unlike consolidations, are sparse enough for vessels and airways to remain visible. Densifications are non-specific findings with several potential causes, such as organising pneumonia, vasculitis or adverse drug reactions. The overall radiological picture, including appearance, location, and the presence or absence of other findings, combined with clinical data, provides the diagnosis of pneumonia. In most patients, radiological signs can be observed within twelve hours after symptom onset and gradually resolve after treatment initiation [210]. There is no pathognomonic radiological pattern in pneumonia associated with specific microbiological aetiology. However, as each microorganism tends to be associated with a certain type of radiological image an indication of the pathogen involved may be provided [211]. Lobar consolidation with densification of an entire or large parts of a lobe is usually seen in bacterial infections caused by *S. pneumoniae, K. pneumoniae* and *L. pneumophila*. Diffuse consolidated parenchymal densification and bronchial wall thickening (so-called bronchopneumonia) suggest an aetiology other than pneumococci, such as *M. pneumoniae, S. aureus* and gram-negative bacteria, while interstitial pneumonia can be seen in viral infections and *M. pneumoniae*. Cavitation is common in infections caused by *S. aureus* and *M. tuberculosis* but also for gram-negative intestinal bacteria such as *K. pneumoniae*. In pneumonia caused by *S. pneumoniae* lobar consolidation is most commonly seen (67% of cases) [212,213] which may involve multiple lobes but rarely cavitates, followed by pleural effusion in 50% of cases [212,214]. Lymphadenopathy is also often seen on CT.

Table 1. Recommended microbiological sampling for adult CAP patients requiring hospitalisation *(C III)*.

	DS-CRB65 0–3	DS-CRB65 4–6	Treatment failure
Blood cultures x II	X	X	X
Sputum culture (or nasopharyngeal culture)	X	X	X
Pneumococcal urinary antigen test	Consider	X	X
Legionella urinary antigen test	Consider	X	Consider
PCR + culture for *Legionella spp*[a]	Consider	X	Consider
PCR for *M. pneumoniae*[b]	Consider	X	X
PCR for influenza, RSV, SARS-CoV-2	Consider[c]	X	X
PCR for other viruses	Consider	Consider	Consider
Diagnostic bronchoscopy	–	Consider	Consider
Extended diagnostic testing[d]	Consider	Consider	Consider

[a]Diagnostics on lower respiratory secretions.
[b]Diagnostics on pharyngeal secretions (flocculated swab, cotton swab or aspirate), sputum or bronchial secretions (BAL, protected brush).
[c]If ongoing community transmission.
[d]Consider e.g. Chlamydia spp, Mycobacterium tuberculosis, B. pertussis, Francisella tularensis, respiratory viruses, fungi.

Chest X-ray

Conventional chest X-ray (frontal and lateral views) is the most established radiological examination for suspected pneumonia [215]. The advantages of this method are its simplicity, availability and low radiation dose (approximately 0.5 mSv). It is also easy to compare over time and between hospitals. The disadvantages include limited sensitivity for detecting pneumonia, especially early in the disease course [216] and a relatively large variability [217,218]. Other underlying lung diseases, such as emphysema, can also complicate the assessment [219,220]. Complete regression is seen in just over half of the patients after two weeks, in two-thirds after four weeks, and changes persist in about 4% after six months. Therefore, 6–8 weeks is a reasonable timeframe for diagnostic imaging follow-up when indicated [221].

Computed tomography (CT)

CT provides higher resolution than chest X-ray [222]. The disadvantages include a higher radiation dose (approximately 5 mSv) [223], less accessibility and more resource consuming. CT is not recommended as a first-line examination but is a valuable complement if suspicion remains despite a negative chest X-ray, if the image is difficult to interpret or if complications occur [216,219,222]. CT is also preferable for immunocompromised patients with suspected opportunistic infections [224].

Ultrasound

Ultrasound examination of the lungs in the emergency department, known as point-of-care ultrasound (POCUS), has become increasingly utilised. According to several studies and meta-analyses, the sensitivity varies between 0.68 – 1.0 compared with chest X-ray and CT, depending on the examiner's experience and the patient population [225–227]. Advantages include simplicity and the ability to visualise signs of pneumonia earlier than conventional chest X-ray [228]. Disadvantages include the inability to examine the entire lung, difficulty in comparing results, and the resource-intensive nature of the examination when performed by a physician. Results also vary greatly between performers. Lung ultrasound is therefore not a routine examination, unlike conventional chest X-ray and CT.

Recommendation

- A radiological examination is recommended for patients requiring inpatient care for CAP *(A II)*.

Antibiotic treatment

S. pneumoniae is both the predominant pathogen and has the highest mortality among common aetiological agents, emphasising that all empirical treatments need to be effective against *S. pneumoniae* [17,23,229]. North American guidelines recommend routine empirical coverage for atypical pathogens in all hospitalised patients with CAP [230] but *Legionella* is an uncommon aetiology for non-severe CAP in Sweden and clinical experience suggests that routine coverage for *M. pneumoniae* and *Chlamydia spp* is unnecessary [231]. This is supported by two randomised studies from the Netherlands and Switzerland [232,233]. In hospitalised patients with pneumonia not requiring ICU care, no difference in mortality between those treated with β- lactam antibiotics alone compared with those treated with β-lactam antibiotics plus macrolide or monotherapy with quinolone was demonstrated [234]. The strengths and weaknesses of these randomised studies have been discussed [235]. A 2024 Greek randomised study demonstrated increased clinical improvement on day 4 among patients receiving clarithromycin plus β-lactam antibiotics compared with β-lactam antibiotics alone, but mortality and aetiological spectra differ from Swedish conditions [234].

Empirical treatment

Table 2 outlines the recommended empirical antibiotic treatment based on severity. For patients with DS-CRB65 scores of 0–1 and 2–3, treatment targets *S. pneumoniae* unless clinical, epidemiological,

laboratory or radiological findings indicated another pathogen. An observational study using data from the Swedish pneumonia registry showed similar mortality in patients with CRB65 scores of 0–2 treated with benzylpenicillin compared to those treated with broader-spectrum antibiotics [236].

Oral antibiotic treatment can be considered for hospitalised patients with non-severe pneumonia (DS-CRB65 0–1 points) and normal intestinal absorption [237–239]. In Sweden, penicillin V is commonly used for non-severe pneumonia [231,240–243], but requires a clinically stable patient and that bacterial culture is performed to guide potential therapy failure. Treatment rationale includes ecological considerations, side-effect profiles, and the favourable resistance situation for *S. pneumoniae* in Sweden [101,243–247]. Amoxicillin is often recommended internationally for CAP due to its higher absorption rate, longer half-life, lower protein binding, and efficacy against β-lactam-sensitive *H. influenzae* [35,248]. If *H. influenzae* is suspected, amoxicillin should be considered at a dosage of 750 mg x 3 [249], but it should be noted that the proportion of amoxicillin-resistant isolates is between 25–35% in Sweden. High-dose amoxicillin (750 mg x 3) is generally effective in treating pneumonia caused by *S. pneumoniae* susceptible to increased penicillin exposure (categorized as I) but isolates with a MIC for PcG of 1 mg/L may be resistant. For *S. pneumoniae* with MIC >1 mg/L, both high dose benzylpenicillin and amoxicillin are insufficient. In cases of type 1 penicillin allergy in non-severe pneumonia, oral doxycycline is recommended [250].

If systemic effects are motivated and for patients with uncertain intestinal absorption, antibiotics should be administered parenterally. 2024 EUCAST recommendations for treatment of *S. pneumoniae* benzylpenicillin should be given in 4-dose or 6-dose regimens, even where a 3-dose regimen was previously customary [100]. The dosage is based on Monte Carlo simulations ensuring that the antibiotic concentration remains above the MIC for at least 40% of the time. We recommend empirical treatment with benzylpenicillin 3 g x 3 as the first choice for non-severe CAP which is equivalent to the recommended EUCAST dosage of 0.6 g x 4 for susceptible *S. pneumoniae* isolates. For patients with DS-CRB65 2-3 points, benzylpenicillin in 3 g x 4 is recommended for patients with failing organ function [100].

Clinical experience indicates that benzylpenicillin 3 g x 3 is effective against most *H. influenzae* strains, which is supported by a Swedish retrospective observational study comparing benzylpenicillin with β-lactam antibiotics with a broader spectrum [19]. No significant differences in mortality were observed. In chronic lung disease, an increased incidence of *H. influenzae* infection is seen, but *S. pneumoniae* remains the most common pathogen according to the Swedish pneumonia registry which justifies

Table 2. Recommended empirical antibiotic therapy for adult patients with CAP requiring hospitalisation based on DS-CRB65 score. Dosage recommendations assume normal renal function.

DS-CRB65 0–1	
Standard patient	Benzylpenicillin 3 g x 3 or penicillin V 1 g x 3 (p.o)
Chronic lung disease[a]	Benzylpenicillin 3 g x 4 or amoxicillin 750 mg x 3 (p.o)
Suspicion of atypical aetiology[b]	Erythromycin 1 g x 2 (p.o) or doxycycline 200 mg x 1 day 1 followed by 100 mg x 1 (p.o)
Penicillin allergy (type 1)	Doxycycline 200 mg x 1 day 1 followed by 100 mg x 1 (p.o) or macrolide e.g erythromycin 1g x 2 (p.o)
DS-CRB65 2–3	
Standard patient	Benzylpenicillin 3 g x 3 – 4[c]
Influenza-associated disease	Cefotaxime 2 g x 3
Underlying lung disease[a]	Benzylpenicillin 3 g x 4
Suspicion of atypical aetiology[b]	Add erythromycin 1 g x 3 or doxycycline 200 mg x 1 day 1 followed by 100 mg x 1
Penicillin allergy (type 1)	Cefotaxime 2 g x 3
DS-CRB65 4–6 or ICU care	
Standard patient	Cefotaxime 2 g x 3 + macrolide (e.g erythromycin 1 g x 3) or fluoroquinolone (levofloxacin 750 mg x 1/moxifloxacin 400 mg x 1) or Benzylpenicillin 3 g x 4 + fluoroquinolone (levofloxacin 750 mg x 1/moxifloxacin 400 mg x 1)
Underlying lung disease[a]	Piperacillin/tazobactam 4 g x 4 + macrolide (erythromycin 1 g x 3) or + fluoroquinolone (levofloxacin 750 mg x 1/moxifloxacin 400 mg x 1)
Penicillin allergy (type 1)	Cefotaxime 2 g x 3 + macrolide (erythromycin 1 g x 3) or + fluoroquinolone (levofloxacin 750 mg x 1/moxifloxacin 400 mg x 1)

[a]Patients with COPD or chronic bronchitis.
[b]Suspected infection with M. pneumoniae acquired in Southeast Asia or C. psittaci; Doxycycline 200 mg x 1.
[c]4-dose regimen is recommended for patients with failing organ function.
[d]If known colonisation of P. aeruginosa or H. influenzae with changes in penicillin binding proteins (PBP) or beta-lactamase production: piperacillin/tazobactam 4 g x 4.

empirical treatment with benzylpenicillin 3 g x 4. Piperacillin/tazobactam 4 g x 4 is however recommended to patients with known colonisation with *Pseudomonas aeruginosa* or β-lactamase-producing *H. influenzae*. In cases of epidemiological suspicion of resistant *S. pneumoniae*, empirical treatment with cefotaxime 2 g x 3 is recommended.

For severe pneumonia (DS-CRB65 4–6 points) or ICU care, empirical therapy with β-lactam antibiotics plus macrolide/fluoroquinolone is recommended to cover atypical respiratory pathogens [251,252]. Unlike the 2023 ERS/ESICM/ESCMID/ALAT guidelines, we do not find enough evidence to recommend macrolides over fluoroquinolones due to the lack of high-quality comparative studies [150]. Regardless of severity, previous microbiological findings should guide empirical antibiotic therapy. Some observational studies have shown reduced mortality with β-lactam antibiotics plus macrolide in pneumococcal aetiology, but this has not been confirmed in other studies [235,251,253–258]. We therefore conclude that there is insufficient support to recommend dual therapy in pneumococcal pneumonia.

Observational studies have shown contradictory results regarding the importance of timing in antibiotic treatment for pneumonia [259,260]. A review has indicated a risk of bias in many of these studies [261]. However, well-conducted studies from the UK and the USA [259,262] demonstrate an improved prognosis if antibiotic treatment is initiated within 4 – 8 h upon hospital arrival. We support initiating empirical treatment within 4 – 8 h, except in cases with septic shock when treatment should be started earlier [261]. This allows sufficient time for microbiological sampling and radiological diagnostics. Patients who receive parenteral treatment should be switched to oral preparations after significant clinical improvement [263–267].

Targeted treatment

Once the aetiology has been established, targeted treatment should be administered [180,268,269], see Table 3.

In cases of *H. influenzae* without resistance to β-lactam antibiotics, where the patient has shown rapid clinical improvement with empirical benzylpenicillin treatment, the regimen can be maintained. However, if the treatment response is inadequate, ampicillin is recommended. *H. influenzae* isolates with changes in penicillin binding proteins (PBP) without β-lactamase production should be treated with cefotaxime, but resistance may be difficult to determine why a switch to piperacillin/tazobactam in cases of severe pneumonia with inadequate treatment response may be needed. For verified infections caused by *Legionella spp*, treatment with fluoroquinolone [270–272] or azithromycin is recommended [273,274]. A 2021 meta-analysis found no difference in mortality in patients with *Legionella spp* treated with a macrolide compared to a fluoroquinolone [275].

Duration of treatment

A treatment duration of 5–7 days is recommended for most patients [276–278]. The shorter treatment duration is recommended for patients who quickly become clinically stable and afebrile [279]. A Danish prospective cohort study from 2023 involving over 11,000 patients found no difference in 30-day mortality between 4–7 and 8–14 days of treatment for patients with rapid clinical improvement [280]. Even for patients with severe uncomplicated pneumonia of unknown aetiology, 7 days of treatment is likely sufficient [281]. The treatment duration for *Legionella spp* should however be 10 days [270,271,282] and for *C. psittaci* 10–14 days [283]. For pneumonia caused by *S. aureus* with concomitant bacteraemia, the treatment duration should be at least 2 weeks, with at least 1 week of parenteral therapy [284,285]. In such cases, the presence of other disease manifestations e.g. meningitis, endocarditis or septic embolisation must be considered. In the case of slow treatment response or complications such as empyema or pulmonary abscess the treatment duration is extended and individualised.

Treatment with parenteral cephalosporines, fluoroquinolones and clindamycin has been associated with *Clostridoides difficile* [286] and gram-negative bacteria that produce extended-spectrum β-lactamase (ESBL) [287–291]. Quinolones and cephalosporins have also been linked to increased incidence of methicillin-resistant *S. aureus* (MRSA) [290,292–294]. However, this association has not been observed in Sweden [101]. Given the increase in ESBL-producing bacteria in Sweden, we recommend that

Table 3. Recommended targeted treatment for CAP with known aetiology.

Aetiology	Intravenous treatment	Oral treatment
S. pneumoniae (MIC ≤0.06 mg/L)	Benzylpenicillin 3 g x 3	Penicillin V 1 g x 3
S. pneumoniae (MIC >0.06 ≤ 1 mg/L)	Benzylpenicillin 3 g x 4	According to antibiogram
S. pneumoniae (MIC >1 mg/L)	According to antibiogram	According to antibiogram
H. influenzae (β-lactamase-negative)	Benzylpenicillin 3 g x 4[a] or ampicillin 2 g x 3	Amoxicillin 750 mg x 3
H. influenzae (β-lactamase-positive but PBP3 negative)	Cefotaxime 2 g x 3	Amoxicillin/clavulanic acid 875/125 mg x 3 or doxycycline 200 mg x 1 day 1 followed by 100 mg x 1
H. influenzae (β-lactamase-negative and PBP3 positive)	Piperacillin/tazobactam 4 g x 4	Doxycycline 200 mg x 1 day 1 followed by 100 mg x 1 or Amoxicillin 750 mg x 3
H. influenzae (β-lactamase-positive and PBP3 positive)	Piperacillin/tazobactam 4 g x 4	Doxycycline 200 mg x 1 day 1 followed by 100 mg x 1
S. aureus (methicillin-susceptible)	Cloxacillin 2 g x 4	Flucloxacillin 1 g x 3
MRSA	Linezolid 600 mg x 2	Linezolid 600 mg x 2[b]
Legionella spp	Levofloxacin 750 mg x 1	Levofloxacin 750 mg x 1 alt Moxifloxacin 400 mg x 1 alt Azithromycin 500 mg x 1
M. pneumoniae and *C. pneumoniae*	Erythromycin 1 g x 3 or doxycycline 200 mg x 1 day 1 followed by 100 mg x 1	Erythromycin 1 g x 2 or doxycycline 200 mg x 1 day 1 followed by 100 mg x 1
C. psittaci	Doxycycline 200 mg x 1	Doxycycline 200 mg x 1

[a]Maintain if clinical improvement, otherwise switch to ampicillin.
[b]Clindamycin 300 mg x 3 or trimethoprim-sulfamethoxazole 160/800 mg x 2 may be considered for sensitive strains and mild disease.

cephalosporins and quinolones primarily are reserved for patients with severe pneumonia and for those who have failed on initial empirical treatment.

Recommendation

- All empirical antibiotic treatment should be effective against *S. pneumoniae (A II)*.
- Treatment should be initiated parenterally if severe disease or if intestinal absorption is uncertain *(A II)*.
- Routine coverage of atypical pathogens is only indicated in severe disease (*C II)*.
- If there is clear clinical improvement, a switch to oral treatment can be made *(A II)*.
- If the aetiology is known, antibiotic treatment should be targeted *(A II)*.

Antiviral therapy

Influenza

Influenza is a prevalent and likely underdiagnosed cause of pneumonia [295]. Antiviral therapy shortens the duration of the illness, reduces the risk of complications, death and the need for antibiotic treatment [296–300]. Risk groups for severe influenza encompass individuals >65 years, pregnant women or <14 days postpartum, and individuals with chronic cardiovascular or pulmonary disease, hepatic or kidney failure and diabetes mellitus. Additional conditions warranting antiviral therapy include compromised pulmonary function, diminished cough efficacy, secretion retention (e.g. extreme obesity, neuromuscular disorders or multiple disabilities) and severely immunocompromised due to disease or treatment [98].

A meta-analysis of randomised trials published in 2015 [301] demonstrated that treatment with neuraminidase inhibitors reduce the duration of the disease and the incidence of influenza-related complications. This analysis included only patients with confirmed influenza virus infection [301]. These findings contrast with a prior Cochrane analysis that included patients irrespective of whether influenza or another respiratory infection was the causative agent of the symptoms [302]. Most guidelines recommend initiation within 48 h of symptom onset to achieve optimal efficacy; however, reduced mortality has been observed with treatment initiation up to 5 days after symptom onset [297,303]. For patients with severe influenza necessitating hospitalisation, antiviral treatment should be considered even after several days of illness [98]. There is likely an additional benefit in reducing healthcare-associated transmission [304,305]. A 2024 review identified eight randomised controlled trials (including a total of 1424

patients) evaluating antiviral therapy in hospitalised patients with severe influenza. The results indicated a positive effect in terms of reduced hospital stay, although the strength of evidence was low [306].

The most used antiviral is the neuraminidase inhibitor oseltamivir. Oral absorption is effective even in severely ill patients and the capsule can also be opened, dissolved and administered by nasogastric tube. The dosage is 75 mg x 2 but should be reduced in case of impaired renal function. Higher dosages does not enhance efficacy [307,308]. Zanamivir is another neuraminidase inhibitor that is administered *via* inhalation at a dosage of 10 mg x 2. This preparation can induce bronchospasm [309] and is thus not recommended for patients with COPD or asthma [305]. Zanamivir is also available as an intravenous preparation with a dosage of 600 mg x 2 with adjustments for renal impairment [307,308]. Studies have indicated a potential advantage of zanamivir over oseltamivir in influenza B infections [310,311].

Baloxavir, approved in 2021 for influenza treatment, inhibits the cap-dependent endonuclease enzyme, and thereby affect viral polymerase activity. Studies have shown comparable efficacy to neuraminidase inhibitors in influenza A infections and equivalent efficacy to oseltamivir in influenza B infections. Baloxavir is administered as a single dose based on weight: 40 mg for individuals under 80 kg and 80 mg for those over 80 kg [299,300]. Baloxavir has demonstrated a faster reduction in viral load compared to neuraminidase inhibitors but also an increased incidence of resistance (approximately 10% of treated patients). The clinical significance of these findings remains unclear [299]. Baloxavir could be an alternative for patients with compliance or swallowing difficulties. Nausea is a common side effect of oseltamivir that can be reduced by concomitant food intake.

PCR testing of respiratory secretions for influenza virus is recommended for accurate diagnosis. If antiviral therapy has been initiated, it can generally be discontinued if the test is negative. In severe cases, bronchoscopy/BAL testing should also be performed. It is important that complementary bacterial culture is performed in hospitalised influenza patients. The most common secondary bacterial infection are *S. pneumoniae, H. influenzae* and *S. aureus* [312]. The Swedish Medical Products Agency recommends oseltamivir prophylaxis (75 mg once daily) for exposed individuals at risk of severe influenza [98]. A 2024 review (including 33 studies and over 19,000 individuals) supports post-exposure prophylaxis for at-risk individuals [313]. For pregnant women, post-exposure prophylaxis is recommended if additional risk factors are present; otherwise, early treatment is advised. The duration for prophylaxis is 10 days. Pre-exposure prophylaxis for extended periods (seasonal prophylaxis) may be relevant for individuals at very high risk of severe influenza.

Additional antivirals

Nucleoside analogue ribavirin has a good effect *in vitro* against RSV infection and has previously been used in trials but is no longer available. A monoclonal antibody for RSV prophylaxis, palivizumab, has been shown to reduce the risk of hospitalisation for young children at high risk of severe RSV infection (premature infants with pulmonary dysplasia) but needs to be given once/month. In 2022, the antibody nirsevimab was approved for newborn/infants a is administered only once/season. The effect is comparable and reduces the risk of serious illness/hospitalisation. Additional monoclonal antibodies are under development. There is currently no established treatment for RSV infection in adults.

Recommendation

- Administer antiviral therapy to individuals at risk for severe illness with suspected/confirmed influenza, regardless of vaccination status if early in the disease course *(AI)*.
- Offer post-exposure prophylaxis for individuals with high risk of severe influenza regardless of vaccination status *(B II)*.

Steroids

The efficacy of steroid therapy in severe CAP has been a topic of extensive discussion. Historically, the scientific evidence supporting the use of steroids has been insufficient due to significant heterogeneity among studies, including variations in inclusion and exclusion criteria and treatment protocols. However,

recent well-conducted studies have provided a basis for recommending steroids in severe CAP under specific conditions. In 2015, two randomised controlled trials demonstrated benefits of steroid therapy [314,315]. One study [314] reported a shorter time to clinical stability in the steroid-treated group, while the other [315] observed a positive 'composite outcome'. Neither found a difference in mortality. A large Japanese study on patients with severe CAP requiring ventilatory support found significantly lower mortality in steroid-treated patients with septic shock, but no benefit in those without septic shock [316]. In a 2022 randomised controlled trial of 584 patients by Meduri et al. was 40 mg of methylprednisolone or placebo administered for 7 days [317]. The study was terminated early due to low enrolment showing no difference in 60-day mortality or secondary outcomes. In 2023, Dequin et al. published a randomised controlled trial comparing hydrocortisone to placebo in 795 ICU patients with severe CAP [318]. The treatment group, receiving 200 mg of hydrocortisone daily *via* continuous infusion for 4 or 8 days followed by tapering (a total of 8 or 14 days of treatment), exhibited significantly lower mortality, fewer intubations, and delayed vasopressor use. However, a substantial proportion of patients were excluded, including those with septic shock, immunosuppression, aspiration pneumonia, influenza, or ongoing corticosteroid therapy.

Several meta-analyses of randomised trials have aimed to synthesise the scientific literature on adjunct steroid treatment. These analyses have outnumbered original studies on the topic [319]. One meta-analysis demonstrated with high certainty that steroid therapy reduced the time to clinical stability and hospital stay duration [320]. Another found that steroid therapy did not cause major side effects and shortened the time to clinical stability and length of hospital stay [321]. However, no mortality difference was found which is in line with previous systematic reviews [322–324]. One study noted an increased risk of readmission in steroid-treated patients [324]. The most recent Cochrane analysis from 2017 included 17 studies [325] and found significantly lower mortality for patients with severe CAP treated with steroids compared to standard treatment. The 2023 ERS/ESICM/ESCMID/ALAT guidelines [150] recommend corticosteroids for CAP in the presence of septic shock, except in patients with uncontrolled diabetes, ongoing corticosteroid therapy, or infections with SARS-CoV-2, MERS or influenza. Corticosteroid treatment equivalent to hydrocortisone 200 mg/day is likely beneficial if circulatory failure persists despite vasopressor therapy. This is supported by a 2024 French study [326] and a retrospective 2015 Japanese study of ventilated patients with severe CAP [316].

Thus, two large randomised studies published in 2022 and 2023 have attempted to clarify the role of corticosteroid therapy as an adjunct treatment for severe CAP with conflicting results [317,318]. Previous assessment that there is insufficient scientific support to recommend steroids *as standard treatment* remains. However, there is reason to consider steroid therapy for severe CAP patients with high expected mortality; for those requiring ICU care, mechanical ventilation or high-flow therapy with FiO2 $\geq$ 50%. Recommended treatment involves continuous intravenous hydrocortisone 200 mg continuous infusion, or 50 mg x 4, for 4 days, with earlier discontinuation if rapid clinical improvement occurs. Randomised studies on steroid efficacy in pneumonia caused by *M. pneumoniae* are lacking. A 2017 Japanese study showed no reduction in mortality [327]. Large randomised controlled trials on systemic steroid treatment in severe influenza are also lacking. A Cochrane analysis from 2021 [328] indicated increased mortality, but the included studies were of low-quality warranting cautious interpretation.

Recommendation

- Initiate treatment $<$24 hours with intravenous hydrocortisone 200 mg/day (continuous infusion or 50 mg four times daily) for 4 days if all the following criteria are met *(B I)*.
- Adult patient with severe CAP requiring ICU (or equivalent) care regardless of septic shock.
- Need for mechanical ventilation (invasive or non-invasive) or high-flow therapy with FiO2 $\geq$ 50%.

Therapeutic failure

Distinguishing between failure to improve and slow clinical response to therapy is essential. The latter is often due to factors such as impaired absorption, drug interactions, adverse reactions or individual variations. CRP levels often increase during the initial days of treatment despite appropriate therapy, but a

decline is anticipated within three days. A reduction in CRP by more than 50% indicates a high likelihood of an uncomplicated course [329]. PCT levels correlate with therapeutic response and may be valuable for monitoring in severe disease [144]. Early consultation with an infectious disease specialist is advisable if therapeutic failure is suspected. Important differential diagnoses include pulmonary embolism/oedema/haemorrhage, vasculitis, malignancy, cryptogenic organising pneumonia (COP) and eosinophilic pneumonia. Additionally, complications such as empyema, lung abscess, acute respiratory distress syndrome (ARDS), metastatic infection and less common pathogens, such as *Pneumocystis jiroveci, Aspergillus species, Pseudomonas aeruginosa, Mycobacterium tuberculosis, Francisella tularensis*, co-infections or resistant organisms should be considered.

Pleural infections

Pleural fluid accumulation due to pneumonia, so-called parapneumonic effusions (PPE), occurs in 36–57% of hospitalised patients with CAP and may contribute to persistent fever despite adequate treatment [330]. Empyema is characterised by the presence of pus in the pleural cavity [331]. Pleural infections typically progress through distinct stages: (1) sterile uncomplicated PPE, (2) complicated PPE (fibrinopurulent stage), and (3) organised stage.

Aetiology is dominated by Gram-positive aerobic cocci; α-streptococci, *S. pneumoniae*, and *S. aureus*. Most pleural infections are polymicrobial, and a higher mortality has been associated with *S. aureus* and Gram-negative bacteria [332]. Anaerobic bacteria are identified in about 25% of cases. Gram-negative bacteria are less common and primarily observed in patients with concomitant chronic diseases [333]. A Swedish study utilising both conventional culture and PCR for 16S rDNA on pleural fluid identified *Streptococcus viridans* and anaerobic bacteria as the most common pathogens [334,335]. Empirical antibiotic therapy should cover both gram-positive aerobic cocci including *S. aureus* as well as *Enterobacterales* and anaerobic bacteria [333]. Suggested antibiotics are cefotaxime + metronidazole or piperacillin/tazobactam [336]. If the aetiology is identified, targeted antibiotic treatment is given. The duration of treatment has not been studied in detail but is often at least three weeks [331].

PPE often resolves spontaneously, however, distinguishing between effusions that resolve with antibiotic therapy alone and those requiring drainage is clinically challenging. Early diagnostic thoracocentesis is recommended for patients with pleural fluid >10 mm in the lateral position [331]. Thoracocentesis with drain insertion is usually performed under ultrasound guidance to minimise organ perforation risk and enhance the diagnostic yield [337]. For aetiological diagnosis, bacterial-specific PCR for 16S rDNA is recommended alongside with bacterial culture, as patients often have ongoing antibiotic treatment. While awaiting microbiological diagnosis, chemical analysis of the pleural fluid can provide support for assessing whether it is a sterile PPE or a complicated PPE with infection in the pleural space, see Table 4.

When analysing pleural fluid, biomarkers such as pH, lactate dehydrogenase (LD), and glucose have ability to differentiate between uncomplicated and complicated PPE, with pH exhibiting the highest discriminatory power [330]. A pH value <7.2 indicates infection and necessitates drainage. A limitation of pH is the requirement for sampling with a heparinised blood gas syringe and access to a blood gas analyser. If pH analysis is not feasible, LD and glucose analysis is recommended [331]. A 2023 Swedish study of 197 patients with CAP and PPE [338], found that lactate was able to distinguish between infected and uninfected pleural fluid similarly to pH. If biochemical analyses do not indicate infection, drain removal is recommended. In cases of purulent effusion or chemical signs of infection, the drain should remain in place and be flushed regularly with 20-30 ml NaCl every 6 h. For purulent effusion, chemical analysis is unnecessary. A pigtail catheter with a diameter of ≤14 F is generally sufficient for drainage [331,339,340]. A study of 405 patients found no differences in outcomes such as length of hospital stay, radiological improvement, need for surgery, spirometry at three-month follow-up, or mortality regardless of drain size from <10 F to >20 F [341]. However, as in previous studies [342], a difference in pain was seen in relation to the size of the drain. In general, drains with a lumen as small as possible are preferable, but in some cases drains with an active suction system (so-called Bülau-drains) are required.

Table 4. Management of suspected pleural infections based on appearance and analysis of pleural fluid.

Macroscopic appearance	Laboratory analysis	Treatment
Clear	pH $\geq$ 7.2 LD $\leq$ 17 microcat /L Glucose $\geq$ 3.4 mmol /L Lactate $\leq$ 4.9 mmol /L. Negative culture	Consider pleural drainage for symptom relief
Clear or cloudy	pH <7.2 LD >17 microcat /L Glucose <3.4 mmol /L Lactate >4.9 mmol /L. Culture often positive	Pleural drainage
Pus	Culture often positive	Pleural drainage

Pleurolysis

Treatment of pleural empyema with pleurolysis, involving intrapleural instillation of the tissue plasminogen activator Alteplase (Actilyse®) and deoxyribonuclease (DNase, Pulmozyme®) has been investigated in a blinded, placebo-controlled study involving 210 patient and resulted in a reduced volume of pleural fluid, fewer patients requiring surgical intervention and shorter duration of hospital stay [341]. The positive effect was not observed when only one of the components was administered. An observational study of 107 patients treated with the same combination demonstrated similar results [343]. In both studies, Actilyse® 10 mg and Pulmozyme® 5 mg were administered intrapleurally twice daily, up to a maximum of six doses. According to the study protocol, Actilyse® 10 mg diluted in 50 ml of 0.9% NaCl was administered first, followed by 10-25 ml of 0.9% NaCl. The drain was then closed for 40–60 min, after which the fluid was allowed to drain freely for 40–60 min. This procedure was then repeated with Pulmozyme® 5 mg diluted in 50 ml of 0.9% NaCl, followed by 10–25 ml of 0.9% NaCl.

A modified treatment regimen with instillation of 10–16 mg Actilyse® and 5 mg Pulmozyme® once daily for three days has been described in several studies [344–346]. The results from the modified regimen appear comparable [341,343]. However, no prospective studies have compared the outcomes between different regimens. Treatment efficacy should be assessed radiologically and consultation with a thoracic surgeon is recommended in complicated cases [331,347]. Intrapleural fibrinolysis with streptokinase or urokinase alone has been primarily used in cases of pleural fibrous septa where adequate drainage is difficult to achieve [331]. While this regimen facilitates drainage, no reduction in mortality, fewer surgical interventions, shorter hospital stays or improved long-term lung function could be demonstrated in a study involving 434 patients [348]. A randomised controlled pilot study from 2022 did not support steroid treatment in PPE [349].

Recommendation

- Thoracentesis should be considered for patients with pleural fluid >10 mm in the lateral position *(B III)*.
- Diagnostic evaluation of pleural fluid should include bacterial culture and PCR for 16S rDNA *(B II)*.
- Empirical treatment should cover gram-positive aerobic cocci, including *S aureus* and *Enterobacteriales* and anaerobic bacteria *(A II)*.

Lung abscess

Lung abscess is an uncommon condition that may arise as a complication of pneumonia with necrosis of the lung parenchyma [350]. This condition is characterised by a suppurative infection where the destruction of lung tissue results in one or more cavities. Bronchopulmonary fistulas can subsequently lead to gas/fluid levels [351]. A similar condition with multiple small cavities (<2 cm in diameter) is termed necrotising pneumonia [351,352]. Both conditions likely represent the same type of pathological process of progressive inflammation. Concomitant pleural empyema may also be present [353]. The primary predisposing factor for lung abscess is aspiration pneumonia with a large inoculum of virulent

bacteria [352]. Aspiration can occur due to decreased consciousness dysphagia, neuromuscular disorders, intestinal obstruction, tonsillectomy and endotracheal intubation [352]. Other predisposing factors include bronchiectasis, pulmonary infarction, septic embolisation, bronchial obstructions and intra-abdominal infections. The diagnosis is confirmed radiologically, where conventional chest X-ray typically shows a rounded and often cavitating induration with gas/fluid levels but CT scans may discriminate between parenchymal and pleural changes and can identify possible broncho-obstructive processes [352]. Bronchoalveolar lavage (BAL) and sampling with a protected brush is recommended for aetiological diagnosis [352,354] and has good specificity and a low risk of complications. Obstructive processes can also be visualised during bronchoscopy. However, its sensitivity for isolating strictly anaerobic bacteria is limited [355]. Sputum culture has lower specificity. In cases of pleural empyema, pleural fluid culture can aid in aetiological diagnosis, though false-negative results are common. Important differential diagnoses include tuberculosis and malignancy including presence of an underlying carcinoma as a potential predisposing condition.

Aetiology

The microbiological agents usually originate from the upper respiratory tract. Anaerobic bacteria (*Prevotella spp Fusobacterium spp*, *Peptostreptococcus spp,* and *Bacteroides spp)* predominate but aerobic bacteria (*S. aureus, S. pyogenes, K. pneumoniae* and *P. aeruginosa)* may also be involved. Mixed infections are common [352,354,356]. Patients with lung abscess often present with several weeks of symptoms, including general malaise, low-grade fever and productive cough. Weight loss and anaemia are common, and patient history may include excessive alcohol consumption, aspiration tendency or decreased consciousness. Lung abscesses are often located in the posterior segment of the right upper lobe, followed by the apical parts of the lower lobes [357].

Antibiotic treatment

Antibiotic therapy should cover common anaerobic bacteria from the oral flora. Efficacy has been shown with both single-agent benzylpenicillin and combination therapy with clindamycin and benzylpenicillin. Benzylpenicillin was previously the first-line treatment for lung abscess [358]. However, later comparative studies demonstrated that clindamycin is more effective, although therapy failure caused by resistant oral bacteria are reported [359,360]. There is no international consensus regarding empirical antibiotic treatment for lung abscess. Parenteral β-lactam/β-lactamase inhibitors, such as piperacillin/tazobactam followed by oral amoxicillin/clavulanic acid, have been suggested [361,362]. One study showed similar efficacy of moxifloxacin in monotherapy compared to β-lactam/β-lactamase inhibitors [363]. A combination of second- or third-generation cephalosporin + metronidazole/clindamycin has also been recommended [364]. Upon clinical improvement and fever resolution, follow-up oral treatment may include amoxicillin/clavulanic acid, clindamycin or moxifloxacin [361,364]. Once the aetiology is established, treatment should be directed with additional coverage for anaerobes. Treatment duration is individualised based on clinical, laboratory and radiological improvement but should last at least 6–8 weeks [365].

The cure rate for conservative treatment with antibiotics and physiotherapy is 80-90% [366]. Some studies support percutaneous drainage [366–368], though the treatment is controversial due to risks such as bronchopleural fistulation [366,369]. Percutaneous drainage is only recommended in selected cases, with consultation from a thoracic surgeon. Lung tissue resection is rarely indicated except in cases of concomitant malignancy. Endoscopic drainage *via* bronchoscope is sometimes an option, assuming the abscess has sufficient airway connection [366].

Recommendation

- Bronchoscopy/BAL and sampling is recommended for aetiological diagnosis *(A III)*.
- Empirical antibiotic therapy should cover anaerobic bacteria from the oral flora *(A III)*.
- Once the aetiology is established, antibiotic therapy should be directed with additional coverage for anaerobes *(C III)*.

- Treatment duration should be guided by clinical and radiological response but last at least 6 weeks *(C III)*.
- Percutaneous drainage treatment is only appropriate in selected cases due to the risk of complications *(C III)*.

Aspiration pneumonia

There are no established diagnostic criteria for aspiration pneumonia. Diagnosis is typically based on a combination of risk factors for aspiration (e.g. swallowing difficulties, decreased consciousness, or impaired cough reflex) along with clinical and radiological findings. Direct observation of an aspiration event is often absent. This patient group is heterogeneous but is estimated to constitute 5–15% of CAP cases [370]. A 2018 American study involving over 190,000 patients found that aspiration pneumonia had a higher mortality rate (29% vs. 12%) compared to other types of CAP, likely due to a more vulnerable patient cohort [371,372].

Aetiology

Microaspiration of secretions from the upper respiratory tract is common among healthy individuals during sleep and is considered a dominant cause of bacterial pneumonia [373]. The development of clinical infection depends on the volume and content of the aspirate, as well as the host's local and systemic immune defences. For virulent bacteria such as *S. pneumoniae* or *H. Influenzae,* a small inoculum may be sufficient to cause infection. Aspiration of larger volumes of secretions (macro aspiration) from the upper respiratory or gastrointestinal tract into the lungs may contain both anaerobic and aerobic bacteria. Radiological changes are typically observed posteriorly in the upper lobes or upper parts of the lower lobes when aspirated in the supine position, or bilaterally at the bases of the lower lobes when the patient is upright. Previous studies indicated a predominance of anaerobes and less virulent oral bacteria, but streptococci, Gram-negative bacteria, and *S. aureus* are also identified [374,375]. The historical view of the area below the larynx as a sterile environment has also been questioned in recent years [376–378].

Antibiotic treatment

There are few systematic studies of antibiotic on the choice of antibiotic therapy for aspiration pneumonia, as well as the optimal duration of treatment. Initial coverage for anaerobic bacteria in addition to oral flora and for gram-negative enteric bacteria is not routinely recommended by IDSA/ATS [230]. However, in cases of suspected abscess, empyema and poor dental status, broader coverage including anaerobes should be considered. The most common anaerobic bacteria isolated are *Peptostreptococcus spp, Fusobacterium spp* and *Bacteroides spp* [357,379,380]. Anaerobic infections generally develop over days to weeks. Routine coverage for atypical respiratory pathogens is not recommended in mild to moderate community-acquired aspiration pneumonia.

Distinguishing bacterial infection from chemical pneumonitis can be clinically challenging. In cases of acute onset dyspnoea, cough, and hypoxaemia following a known or suspected aspiration event (e.g. gastric acid aspiration during intubation) without radiological findings and with mild to moderate symptoms, observation may be an alternative. Various treatment strategies have been effective for aspiration pneumonia, including piperacillin/tazobactam, moxifloxacin, clindamycin, carbapenems, benzylpenicillin + metronidazole and amoxicillin/clavulanic acid [375,381]. However, supporting evidence for the different treatment regimens is limited [382] and the choice of antibiotic is primarily guided by disease severity. Repeated episodes of aspiration pneumonia can lead to chronic pulmonary damage and fibrosis, why minimisation of risk factors is essential.

Recommendation

- Antibiotic therapy for community-acquired aspiration pneumonia is primarily directed against oral flora, e.g. benzylpenicillin 3 g x 3 *(C III)*.

- Gram-negative coverage may be justified if aspiration of intestinal contents is suspected, e.g. cefotaxime 2 g x 3 or piperacillin/tazobactam 4 g x 3 *(C III)*.
- Routine coverage of anaerobes and atypical bacteria is not recommended *(C III)*.
- Expectancy is an alternative for aspiration with mild symptoms and without radiological findings *(C III)*.

Discharge, follow-up and prevention

Patients hospitalised for pneumonia can typically be discharged after at least 24 h of clinical improvement [264,267]. The following criteria have been proposed: respiratory rate <24/min, saturation >90%, normalised blood pressure and mental status [267]. Patients hospitalised for CAP, especially the elderly, has an increased the risk of cardiovascular and pulmonary complications [16,383,384] and should be offered follow-up. For uncomplicated cases, a telephone contact may be sufficient [385] but more complicated cases should be offered a return visit [16,383,386] and a follow-up chest x-ray after 6–8 weeks. In addition, patients with recurrent pneumonia, immunosuppression or increased risk of underlying malignancy (primarily for smokers >40 years or former smokers >50 years) should undergo follow-up chest x-ray. If persistent symptoms and/or atypical radiological findings, CT and possibly bronchoscopy should be performed [387].

Since CAP is associated with significant morbidity and mortality, preventive measures are important. Smoking is a known risk factor for both pneumonia and invasive pneumococcal disease. Patients with increased risk for severe COVID-19, influenza, pneumonia and RSV infections should be identified and offered vaccination in accordance with national recommendations. Pneumococcal vaccination is advised for specific risk groups, including those with a history of with pneumococcal bacteraemia. In addition, individuals with severe pneumococcal pneumonia may be candidates for vaccination. Two new RSV vaccines were approved in 2023, one of which is also approved for pregnant women. For vaccination recommendations and background information, refer to the Public Health Agency of Sweden [388].

Recommendation (CIII)

- Patients can be discharged after at least 24 hours of clinical improvement/stable vital parameters.
- Follow-up by telephone if uncomplicated disease course, return visit if complicated and referral to primary care if underlying COPD is suspected.
- Follow-up chest X-ray for complicated courses and smokers >40 years old/former smokers >50 years old.
- Provide information on preventive measures (smoking cessation and vaccination).
- Provide vaccination in accordance with the recommendations from the Public Health Agency of Sweden.

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PN collaborates with MSD regarding pneumococcal vaccination for children.

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